

UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2020/2021

JANUARY 2021 TRIMESTER

FINAL ASSESSMENT

UCCD1004 PROGRAMMING CONCEPTS AND PRACTICES

FRIDAY, 30 APRIL 2021

TIME : 3.00 PM – 6.00 PM (3 HOURS)

BACHELOR OF COMPUTER SCIENCE (HONOURS)
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
BUSINESS INFORMATION SYSTEMS
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
INFORMATION SYSTEMS ENGINEERING
BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)
COMPUTER ENGINEERING
BACHOLOER OF INFORMATION TECHNOLOGY (HONOURS)
COMMUNICATIONS AND NETWORKING

Instructions to Candidates:

REMINDER: You are reminded to read and adhere to the Final Assessment Instructions to Candidates that has been made available through the UTAR Portal before the commencement of Final Assessment (FA). The detailed instructions for this FA are as follows:

General

1. This Final Assessment is an Individual, time restricted assessment which consists of FOUR (4) questions. Each question carries 25 marks.
2. You are required to answer **ALL** questions, and submit the **ANSWER SCRIPT** by **6.00 pm, 30 APRIL 2021**.
3. During the period of 3 hours of this FA, the examiner(s) can be reached at:
 - Microsoft Team
 - Email: jstan@utar.edu.my

You may use the above e-platform(s) to check with the examiner(s) if you need any clarification on this FA question paper.

4. You may refer to any books, lecture notes, published materials, online resources, etc when answering the questions. Candidates are reminded that **COPY-AND-PASTE, CONSULTATION, DISCUSSION AND SHARING OF ANSWERS** with others are **STRICTLY PROHIBITED** in this FA.

UCCD1004 PROGRAMMING CONCEPTS AND PRACTICES**Answer Script File**

5. The **answer of each question** should be **labeled** and stated clearly.
6. The answer script **MUST** be a **PDF**, in A4 size format.
Note: Please keep the file size NOT exceeding 10MB per file.
7. Please check your index number generated by the Division of Examinations, Awards, and Scholarships (DEAS). You **MUST** name your answer script file using the following file name for submission:
[Course Code]_FA_[Programme Abbreviation]_[Your Index Number]

For example, if you are from the degree programme IB, and your Index Number is A01234CBIBF, then your answer script should be named as
UCCD1004_FA_IB_A01234CBIBF.pdf

Answer Script Submission

8. Your answer script file has to be submitted to the following **TWO (2) platforms** before the due time/date.
 - **SUBMIT** your answer script to the google form with the link provided as below:
<https://docs.google.com/forms/d/e/1FAIpQLSfphKHWtqxTqi9wVd8uHhZCGbl3s4VmVuxqVvb2lV9jzz2kyA/viewform>
 - Send your answer script to the following Email Account:
Email: **UCCD1004.FICT@utar.edu.my**
Note: For the title of your email, please use the file name of your answer script.
E.g., UCCD1004_FA_[Programme Abbreviation]_[Your Index Number]
9. Please make sure you submit the correct, complete and final version of your answer script and the same copy of answer script is to be submitted to the above platform(s).
10. **ONLY ONE TIME SUBMISSION IS ALLOWED IN WBLE.**
Please make sure you **CHECK** the answer script **MULTIPLE TIMES** before clicking the submit button in **WBLE**. Any request of **RE-SUBMISSION** will be **IGNORED** except valid reason is provided.
11. Please make sure that you **SUBMIT** the **SAME COPY** of answer scripts to both mentioned platforms. The examiner will only pick the **WBLE** submission as the **PRIMARY** source for **MARKING** and the other will be totally **ignored** (**Unless** the file in the **WBLE** submission is **CORRUPTED**).

UCCD1004 PROGRAMMING CONCEPTS AND PRACTICES**Contents of Answer Submission**

12. The **FIRST** page of your submission is the **Final Assessment Cover Page**. You **MUST** use the template given and fill in the following information
 - Your Programme (Abbreviation)
 - Your Index Number
 - Your Name
 - Your Student ID
13. The **second** page of your submission is the **Final Assessment Declaration Statement**. You **MUST** use the template given, and digitally sign on the form to indicate the authenticity of your submitted work is without plagiarism.
14. Each question should be answered starting on a new page.
15. For answer scripts that have text-based answers only, all texts **MUST** be typed and recommended using Times New Roman characters with font size 12, with proper spacing and alignment.
16. Please include a page number on each and every page of your answer script. Ensure that each page of answer scripts is in sequence prior to online submission.

WARNING OF PLAGIARISM

17. Answer scripts shall be uploaded to Turnitin to check for the originality of submitted answers. In the case of suspected plagiarism, the evidence will be submitted to the Examination Disciplinary Committee of the University. Disciplinary action shall be taken against any candidate who is found to have plagiarized in the answer submitted. Hence, candidates are reminded to abide by all University Rules and Regulations and any instructions/guidelines relating to examinations/assessments.

UCCD1004 PROGRAMMING CONCEPTS AND PRACTICES

Q1. (a) In a C++ program, variables are declared as below:

```
int a = 100, b = 10, c = 6;
double d = 5.5, e = 2.5;
```

With the declared variables,

- (i) Define the library that is able to produce the values of (iii) and (iv) as shown in the output segment below. (1 mark)

```
Welcome to Division Section
(i)   Division of a/b = 10
(ii)  Division of a/c = 16
(iii) Division of c/b = 1.00
(iv)  Division of b/a = 0.10
```

- (ii) Write the codes that able to COUT the exact outputs as shown in the segment above by using the proper stream manipulators. (9 marks)
- (iii) Write the codes that able to COUT the outputs as shown below by replacing the ? with the correct operator.

The output of $b \text{ ? } d \text{ ? } e = 137.50$

The output of $((b \text{ ? } c) \text{ ? } (d \text{ ? } e)) = 1$

(5 marks)

- (b) Write a program that let user to key in two integer numbers and perform calculation by selecting one of the math's operators (+, -, * and /) as shown in Figure 1.1. *Note: any number divided by zero is invalid.

```
Key in two numbers: 10 10
1. +
2. -
3. *
4. /
Select operator: 1

10 + 10 = 20
```

```
Key in two numbers: 5 7
1. +
2. -
3. *
4. /
Select operator: 2

5 - 7 = -2
```

```
Key in two numbers: 14 25
1. +
2. -
3. *
4. /
Select operator: 3

14 * 25 = 350
```

```
Key in two numbers: 9 0
1. +
2. -
3. *
4. /
Select operator: 4

Invalid operation!
```

Figure 1.1: Four sample outputs of basic mathematical operations

(10 marks)

[Total : 25 marks]

UCCD1004 PROGRAMMING CONCEPTS AND PRACTICES

Q2. (a) Given that

```
int x = 135, y = 38, z = 5;
z = ++x - y-- / --z;
```

evaluate the following comparison expression (within **if()**) to TRUE or FALSE. You need to show every step involves. Treat each of the statement separately.

(i) **if** ($z/3 \leq y \ \&\& \ y < z$) (3 marks)

(ii) **if** ($!(--z - 125) \ || \ (--y > x-- \% 2)$) (3 marks)

(iii) **if** ($!(y < 38 \ || \ 0)$) (3 marks)

(b) Given the following code fragment in Figure 2.1:

```
int a, b = 3 , c;
cout<<"Please enter value a and c:"<<endl;
cin>>a>>c;
c %= (++a - b--);
if (a == 6)
if (c == 2)
cout<< c-- <<"\n";
cout<< ++c <<endl;
else
cout<< --c <<endl;
cout<< c++ <<endl;
```

Figure 2.1 C++ Code Fragment

Insert curly braces to proper position in the codes above to get the following output below:

(i) Assuming a=5 and c=5, output wanted is: (3 marks)

2
2

(ii) Assuming a=5 and c=5, output wanted is: (3 marks)

2
2
2

UCCD1004 PROGRAMMING CONCEPTS AND PRACTICES**Q2. (Continued)**

- (c) Table 1 below shows the charges for courier service of J&J courier company based on the weight of parcel. Write a complete C++ program which allows admin to calculate the total prices that a customer need to pay. A customer might have more than one parcel to deliver. A sample output is shown in Figure 2.2. ***Note:** Input validation is not required.

Table 1. Charges of Courier Service

Parcel Weight	Peninsular Malaysia	East Malaysia
1kg and below	8.50	12.50
1kg<x<=2kg	14.00	20.00
2kg<x<=3kg	24.00	32.00
Above 3kg additional per kg	3.00	

Pick your delivery region:
1. Peninsular Malaysia
2. East Malaysia
1
Please key in the weight of the parcel in kg: 4.56
Do you have another parcel?
Y
Pick your delivery region:
1. Peninsular Malaysia
2. East Malaysia
2
Please key in the weight of the parcel in kg: 2.00
Do you have another parcel?
N
Total payment: RM 50.00.

(Hint: Y = Yes; N=No)

Figure 2.2. Sample Output

(10 marks)
[Total : 25 marks]

UCCD1004 PROGRAMMING CONCEPTS AND PRACTICES

- Q3. A Malaysian is required to register himself/herself with Election Commission of Malaysia (SPR) in order to vote in the coming election. The eligibility of age to vote in Malaysia is 21 years old and above. In order to check whether he/she is eligible or not, name and Identity Card (IC) numbers are required as input into the program as shown below:

```
Name: Rambo Hong
IC No (without hypen): 650808086767
Rambo Hong is 56 years old.
Rambo Hong is eligible to vote!
Cont (1-yes, 2-no)? 1

Name: Tan Sui Sui
IC No (without hypen): 101010147878
Tan Sui Sui still has 11 years to reach 21 years old.
Tan Sui Sui is not eligible to vote!
Cont (1-yes, 2-no)? 1

Name: Kong Min Min
IC No (without hypen): 990929085566
Kong Min Min is 22 years old.
Kong Min Min is eligible to vote!
Cont (1-yes, 2-no)? 2
```

Important Notes:

***Format of a Malaysian IC, e.g. 990929085566 (as shown in the example):
 990929 → date of birth which is 29th of September 1999
 08 → state of birth
 5566 → index number

Assuming *string* name and *string* IC are declared in the **MAIN** function as shown below:

```
int main() {
    string name, IC;
    int cont = 1, year;
    bool check;

    do {
        //(A) Key in name and IC
        //(B) Check the characters in IC
        if (checkCharacter(IC)) {
            //(C) Extract the year of birth from IC
            year = convertYearofBirth(IC);
            //(D) Check eligibility of voting
            checkEligibility(name, year);
        }
        else
            cout << "Please key in the correct IC no!" << endl;

        cout << "Cont (1-yes, 2-no)? ";
        cin >> cont;
        cin.ignore();
    } while (cont == 1);
    return 0;
}
```

UCCD1004 PROGRAMMING CONCEPTS AND PRACTICES**Q3. (Continued)**

The following are the descriptions for A, B, C and D sections in **MAIN** function including the **THREE (3)** functions used.

- A. Inputs such as name and Identity Card (IC) numbers are required to be keyed in by the user in order to determine whether he/she is eligible to vote or not.
- B. ***bool checkCharacter(string IC)***
Check for any non-numeric inputs of the IC. **Exit loop** to proceed if the inputs **contain non-numeric input**.
- C. ***int convertYearofBirth(string IC)***
Extract the year (1st character and 2nd character) from the IC and return to the **MAIN** function with a proper format (***Note:** `stoi()` can be used to convert from **string** to **int**).
E.g., The **proper format** of 10 is 2010.
- D. ***void checkEligibility(string name, int year)***
Check the voting eligibility of a Malaysian using the years obtained from C (assume the election year is 2021).

Based on the definition above, answer the following questions by providing C++ codes.

- (a) Write the function prototypes of *checkCharacter*, *convertYearofBirth* and *checkEligibility* functions. (3 marks)
- (b) Write the codes that receive the name and the IC as the inputs as describe in the **section A**. (2 marks)
- (c) Write function definition for *checkCharacter* to ensure users key in only numeric input. (6 marks)
- (d) Write function definition for *convertYearofBirth* to obtain the proper format of the years. (8 marks)
- (e) Write function definition for *checkEligibility* to ensure all the voters are 21 years old and above. (6 marks)

[Total : 25 marks]

UCCD1004 PROGRAMMING CONCEPTS AND PRACTICES

Q4. (a) Given a C++ structure as below:

```
struct Address{
    int houseNumber;
    char streetName[100];
    int postcode;
    string city, state;
}address;
```

By using C++ define a struct called **PaintingService** that consists of four variables as follows: *the type of house as int, verify whether the house is full of furniture or not as char, price of services as double* and nested structure of *address*. (5 marks)

(b) In C++, declare an array of **PaintingService** type called **PaintingList** that is defined in Q4(a) with 100 elements. (3 marks)

(c) Write a function named **void PaintingInfo(PaintService PaintingList[])** to get the information about type of house, house address and verification of the house furniture. The sample of the output is shown in Figure 4.1.

```
Please key in your house type: 1-Flat 2-Apartment 3-Condominium 4-
Single Storey 5-Double Storey
4
Please key in your house number: 24
Please key in your street name: Jalan Universiti
Please key in your postcode: 31900
Please key in your city: Kampar
Please key in your state: Perak
Is your house full of furniture and needs to be shifted? Y=Yes
N=No: Y
Do you want to key in another painting service info? Y=Yes N=No: Y

Please key in your house type: 1-Flat 2-Apartment 3-Condominium 4-
Single Storey 5-Double Storey
2
Please key in your house number: 23-8-78
Please key in your street name: Jalan Masjid
Please key in your postcode: 31900
Please key in your city: Kampar
Please key in your state: Perak
Is your house full of furniture and needs to be shifted? Y=Yes
N=No: Y
Do you want to key in another painting service info? Y=Yes N=No: N
```

(Hint: Y = Yes; N=No)

Figure 4.1 Sample output

(7 marks)

UCCD1004 PROGRAMMING CONCEPTS AND PRACTICES**Q4. (Continued)**

- (d) Write a function named ***void TotalChargedEstimation(PaintingService PaintingList[])*** to calculate the estimated charges of each painting services and also total charges from all painting services. The estimated price is listed as in Table 2. Then, write the painting services details to the output file of *painting.txt* as in Figure 4.2.

Table 2 Estimated Price List for Painting Service

Type of House	Estimated Price (RM)
Flat	1700
Apartment	2000
Condominium	2200
Single Storey	2500
Double Storey	3300

Noted: Additional RM250 if the house is full of furniture and needs to be shifted.

Single Storey 24, Jalan Universiti, 31900, Kampar, Perak. House with furniture: Y Estimated charges: 2750 Apartment 23-8-78, Jalan Masjid, 31900, Kampar, Perak. House with furniture: N Estimated charges: 2000 Total estimated charges from painting services is 4750.
--

Figure 4.2 Output file of painting.txt

(10 marks)

[Total : 25 marks]